

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.7199505905549863, median 0.49906905980877236, std: 1.0243295024352335
Gyroscope error (imu0): mean 0.9046893974219513, median 0.7284992540497243, std: 0.9933800037716114
Accelerometer error (imu0): mean 0.8992836308610557, median 0.7807820995492373, std: 0.6872282196041961

Residuals

Reprojection error (cam0) [px]: mean 0.7199505905549863, median 0.49906905980877236, std: 1.0243295024352335
Gyroscope error (imu0) [rad/s]: mean 0.0020576238876432677, median 0.0016568973523228814, std: 0.002259341638237742
Accelerometer error (imu0) [m/s^2]: mean 0.04513688434418548, median 0.03918905016832151, std: 0.03449338962906994

Transformation (cam0):

T_ci: (imu0 to cam0):

[[0.02644196 0.99608247 0.08438321 0.0324735]
[0.99885381 -0.02295747 -0.04200033 0.03381097]
[-0.03989857 0.08539706 -0.99554781 -0.0458736]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[0.02644196 0.99885381 -0.03989857 -0.03646117]
[0.99608247 -0.02295747 0.08539706 -0.0276526]
[0.08438321 -0.04200033 -0.99554781 -0.04698951]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.10104528268444947

Gravity vector in target coords: [m/s^2]

[-0.22216104 -9.80373478 0.07649636]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [156.65060758519883, 160.18419084516256]

Principal point: [320.49350061169537, 240.12256798837313]

Distortion model: radtan

Distortion coefficients: [-0.25251023886519247, 0.04502550993293753, 0.0004831412405448939,
-0.000760323568641316]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.083 [m]

Spacing 0.024900000000000002 [m]

IMU configuration

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IMU0:

Model: scale-misalignment

Update rate: 200.0

Accelerometer:

Noise density: 0.003549113528381098

Noise density (discrete): 0.050192044862383774

Random walk: 0.0004892041073529009

Gyroscope:

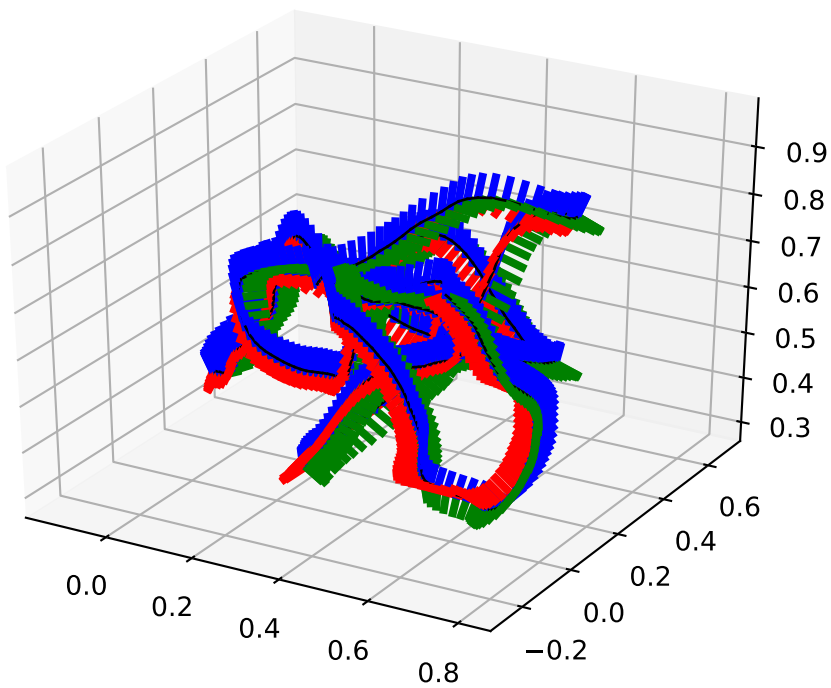
Noise density: 0.00016082423517177371

Noise density (discrete): 0.002274398145382025

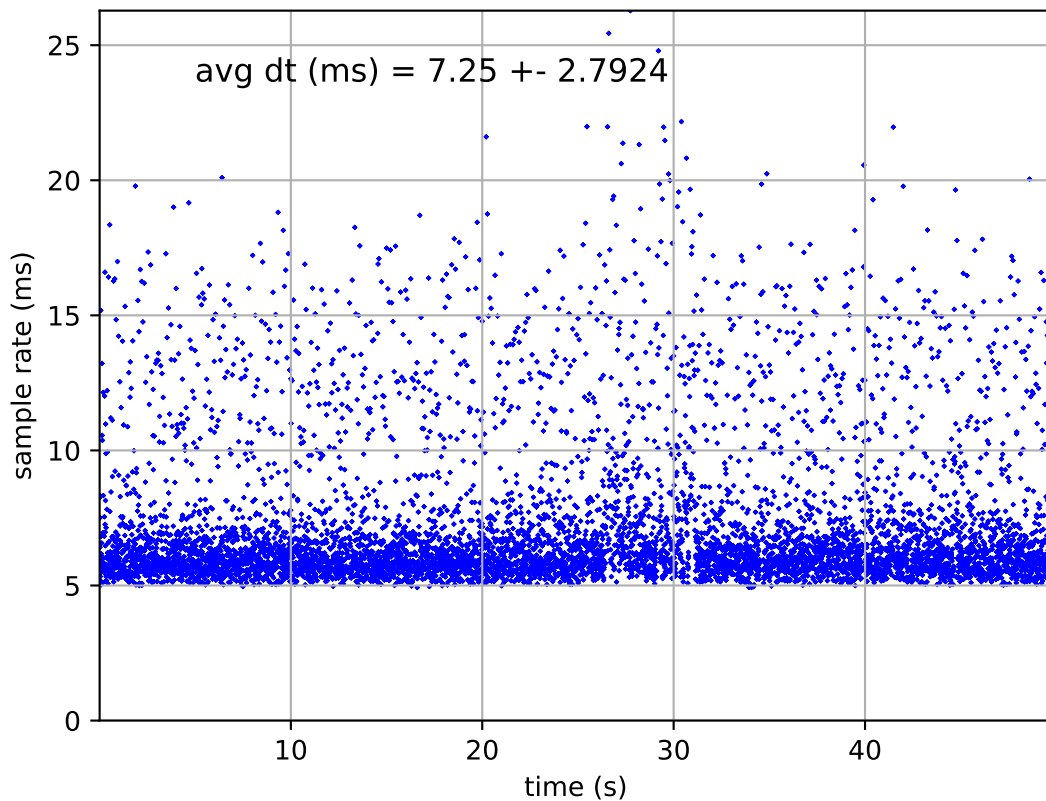
Random walk: 1.3738919624319823e-05

T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
Gyroscope:
M:
[[1.0416442 0. 0.]
[-0.03756138 1.04250564 0.]
[-0.04094375 0.03167247 0.99906649]]
A [(rad/s)/(m/s^2)]:
[[-0.00204612 -0.00008656 0.00307623]
[0.00291717 0.00068082 -0.00261978]
[0.00009607 -0.00007133 -0.0006414]]
C_gyro_i:
[[0.99872175 -0.03135954 -0.03964153]
[0.0326832 0.99891434 0.03319567]
[0.03855749 -0.03444885 0.9986624]]
Accelerometer:
M:
[[1.03474353 0. 0.]
[0.00041138 1.01362266 0.]
[0.01845511 -0.0328485 1.05451431]]

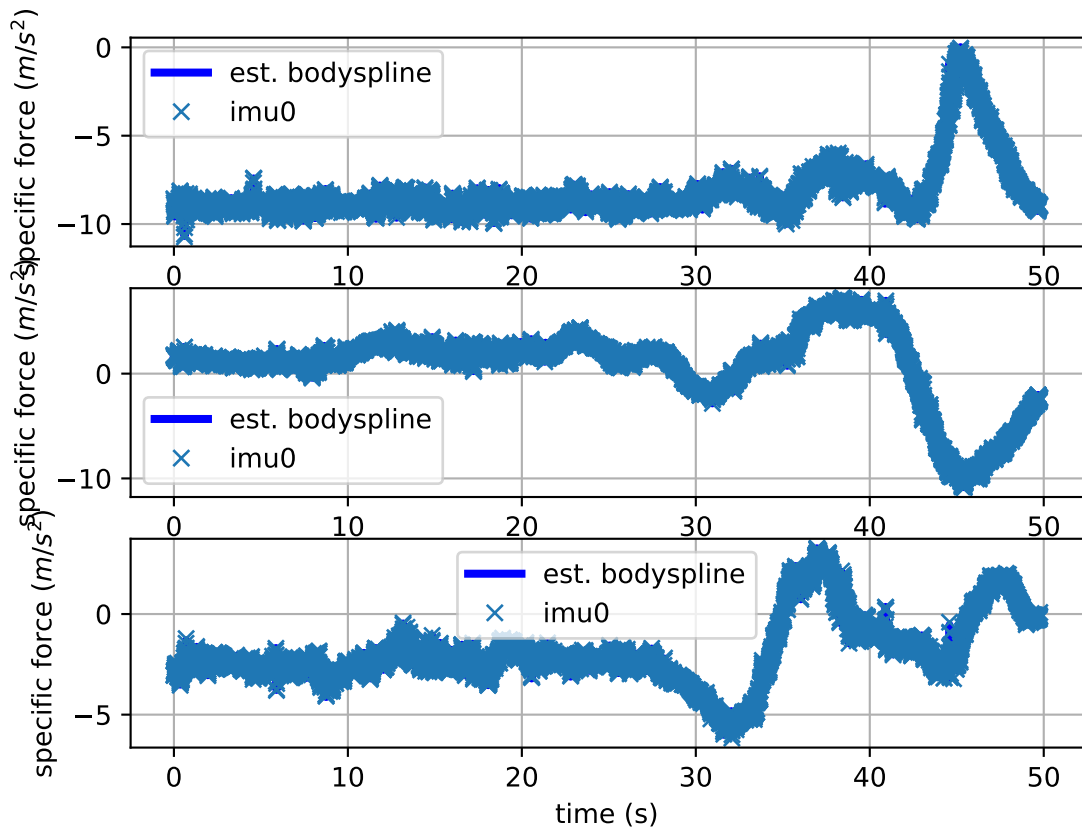
imu0: estimated poses



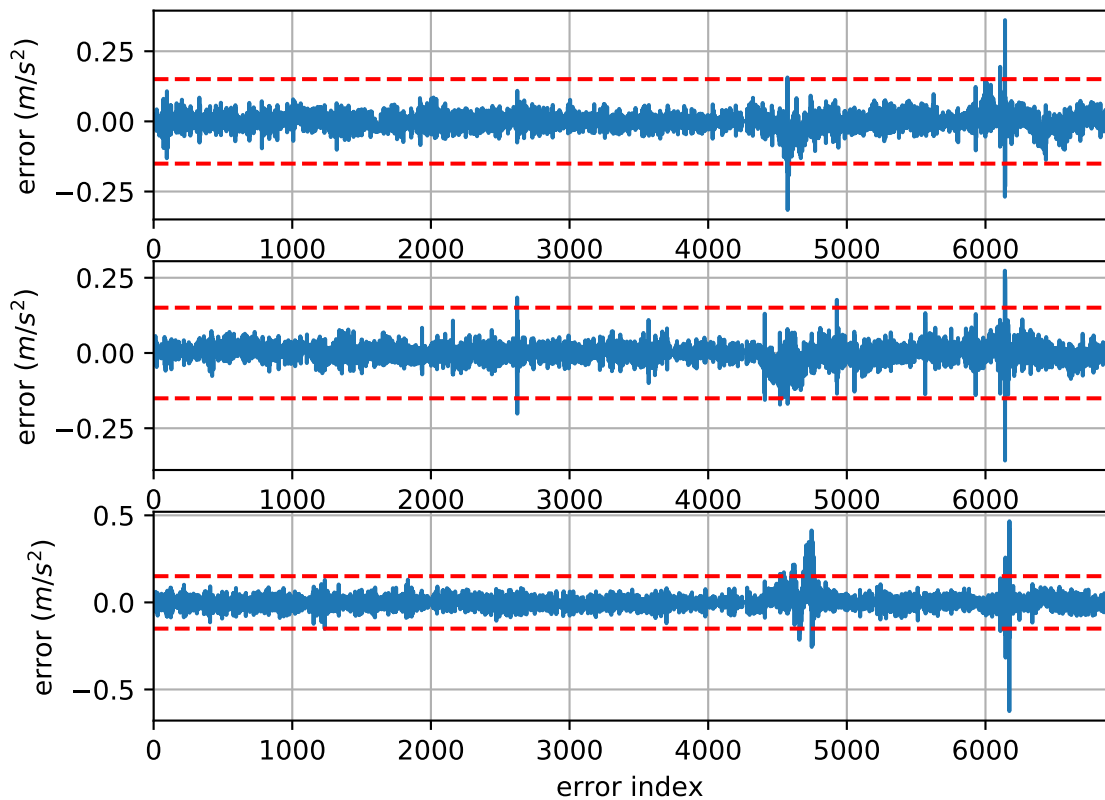
imu0: sample inertial rate



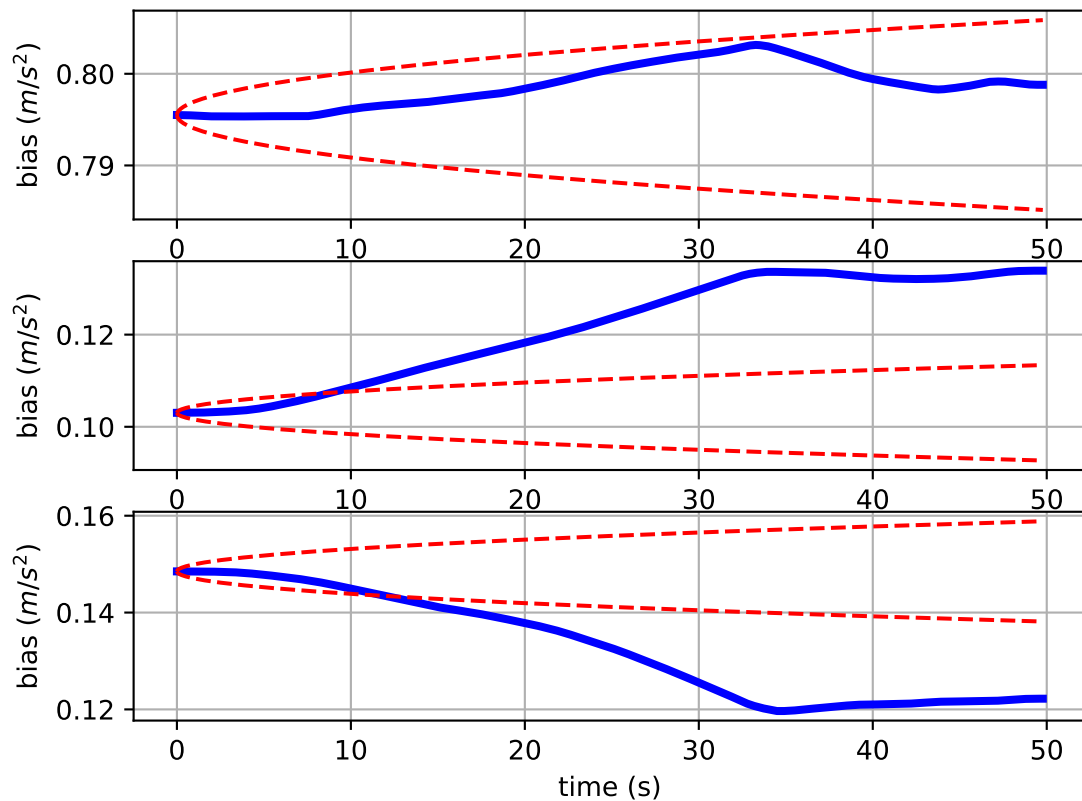
Comparison of predicted and measured specific force (imu0 frame)



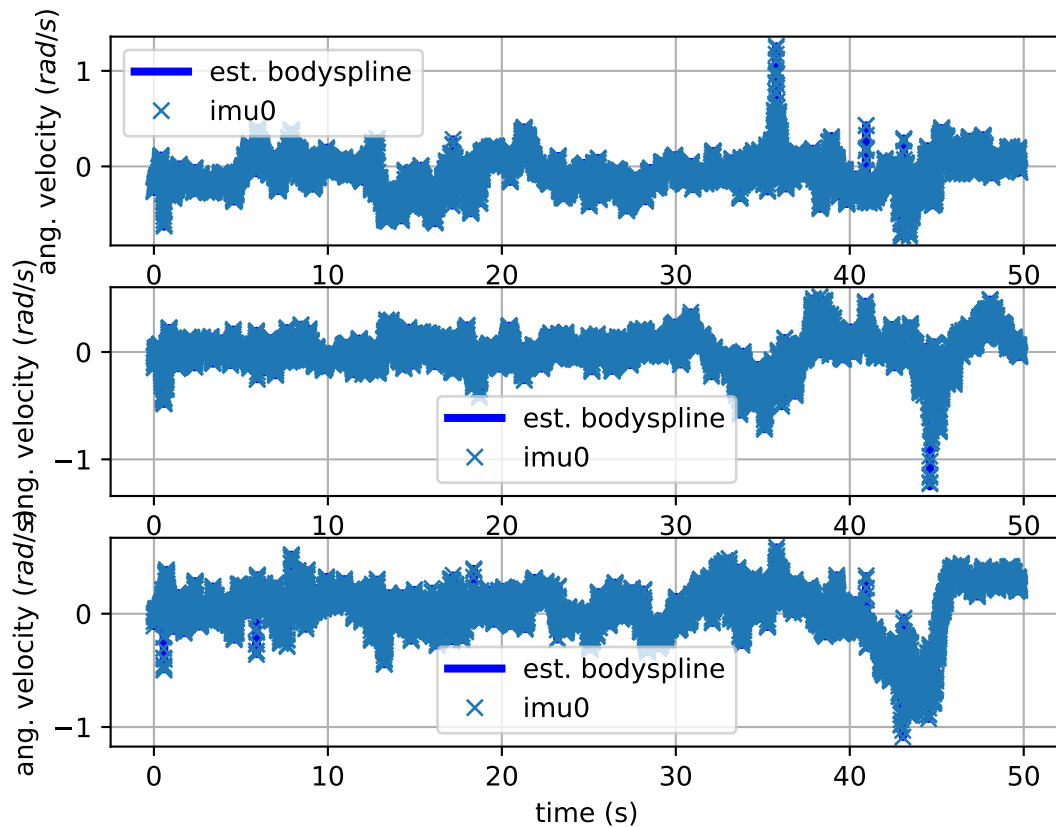
imu0: acceleration error



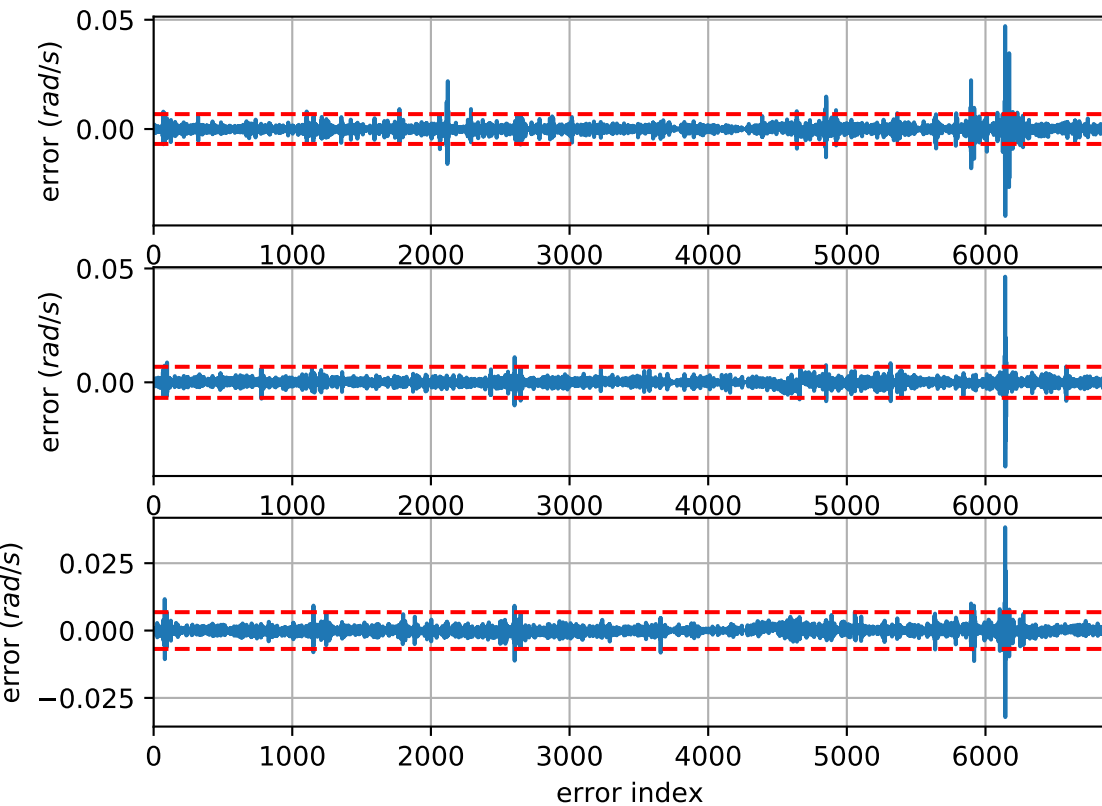
imu0: estimated accelerometer bias (imu frame)



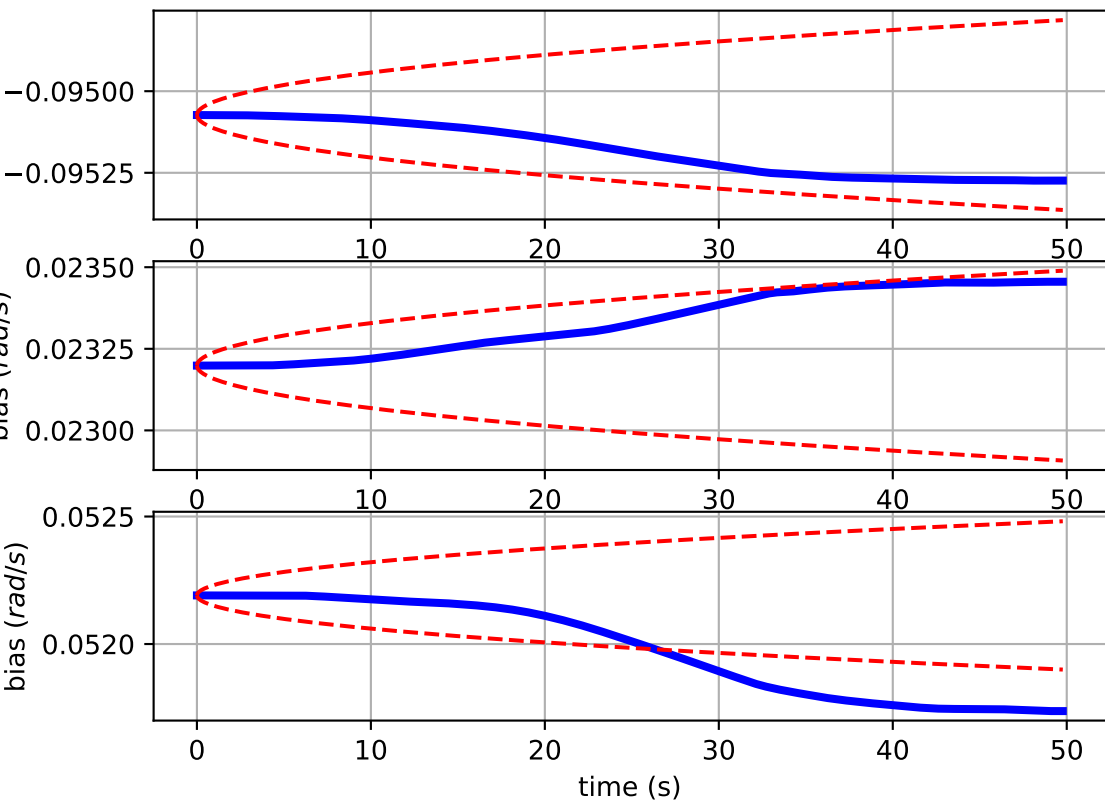
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

